

DESIGN AND FABRICATION OF SUSTAINABLE SMART WATER FILLING SYSTEM USING SENSOR BASED AUTOMATION

PROTOTYPE LAB

Submitted by

MONISH R	(727723EUMT082)
NAVITH SANJAY NAGENDRAN	(727723EUMT087)
SANTHOSH KUMAR R	(727723EUMT120)
SRI PRENESH V	(727723EUMT129)

In partial fulfillment forthe award of the degree of

BACHELOR OF ENGINEERING

IN

MECHATRONICS ENGINEERING



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008

Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

OCTOBER 2025



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

DESIGN AND FABRICATION OF SUSTAINABLE SMART WATER FILLING SYSTEM USING SENSOR BASED AUTOMATION

A PROJECT REPORT

Submitted by

MONISH R (727723EUMT082)

NAVITH SANJAY NAGENDRAN (727723EUMT087)

SANTHOSH KUMAR R (727723EUMT120)

SRI PRENESH V (727723EUMT129)

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

MECHATRONICS ENGINEERING

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

**An Autonomous Institution | Approved by AICTE | Affiliated to Anna
University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008.**

OCTOBER 2025



SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals are a collection of 17 global goals designed to blue print to achieve a better and more sustainable future for all. The SDGs, set in 2015 by the United Nations General Assembly and intended to be achieved by the year 2030, In 2015, 195 nations agreed as a blue print that they can change the world for the better. The project is based on one of the 17 goals.

Questions	Answer Samples
Which SDGs does the project directly address?	This project supports SDGs 6, 7, 9, 11, and 13 through smart water management, energy efficiency, automation and climate resilience.
What strategies or actions are being implemented to achieve these goals?	The project meets SDGs 6, 7, 9, 11, and 13 via water level monitoring, temperature control, energy efficiency, alerts system, and automation.
How is progress measured and reported in relation to the SDGs? How were these goals identified as relevant to the project's objectives?	Progress is measured by water savings, energy reduction, and system performance. These goals were identified based on the project's focus on sustainable water use, efficient energy management, and improved resilience.
Are there any partnerships or collaborations in place to enhance this impact?	Partnerships with academic mentors, technical guides, and component providers enhance water, energy, and infrastructure expertise.



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore – 641008

Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

BONAFIDE CERTIFICATE

Certified that this Mini project report “ **DESIGN AND FABRICATION OF SUSTAINABLESMART WATER FILLING SYSTEM USING SENSOR BASED AUTOMATION** ” is the bonafide work of “ **NAVITH SANJAY NAGENDRAN (727723EUMT087), MONISH R (727723EUMT082), SRI PRENESH V (727723EUMT129), SANTHOSH KUMAR R (727723EUMT120)** ” who carried out the Project work under my supervision.

SIGNATURE

Dr. M. LYDIA

HEAD OF THE DEPARTMENT,

Department of Mechatronics

Engineering,

Sri Krishna College of Engineering
and Technology, Coimbatore -641008.

SIGNATURE Dr. M.

BHUVANESWARI, ASSISTANT

SUPERVISOR, Department of

Mechatronics Engineering, Sri Krishna

College of Engineering

and Technology, Coimbatore -641008.

Submitted for the Prototype lab viva-voce examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

At this juncture, we take the opportunity to convey our sincere thanks and gratitude to the management of the college for providing all the facilities to us. We wish to convey our gratitude to our college Principal Dr . K . PORKUMARAN, for supporting us to do our project and offering adequate duration to complete our project. We would like to express our grateful thanks to Dr. M. Lydia, Head of the Department, Department of Mechatronics Engineering for her encouragement and valuable guidance on this project. We extend my gratitude to our beloved guide Dr. M. BHUVANESWARI, Assosicate professor, Department of Mechatronics Engineering for her constant support and immense help at all stages of the project.

ABSTRACT

The Sustainable Smart Water Filling System Using Sensor-Based Automation is an innovative and eco-friendly solution designed to optimize water usage, energy consumption, and user convenience through intelligent automation. The system integrates a combination of sensors, actuators, and control modules to automatically manage water filling, heating, and monitoring processes without manual intervention. An ultrasonic sensor continuously monitors the water level to prevent overflow and dry running of the pump, while a temperature sensor ensures precise control over the water heating process. The microcontroller acts as the central control unit, processing real-time sensor data and operating relays to automate the pump and heater. A GSM module enables instant communication by sending alerts to the user in case of abnormal conditions such as low water levels or overheating. In addition, motor drivers facilitate mechanical operations, and status indicators provide visual feedback for ease of monitoring.

The entire system is powered through an SMPS and buck converter setup, ensuring stable voltage regulation and energy efficiency. Designed with reliability, sustainability, and affordability in mind, the system is suitable for residential, commercial, and small-scale industrial applications. By integrating automation and intelligent control, this smart water management solution minimizes resource wastage, promotes energy conservation, and contributes to the broader goal of sustainable living.

TABLE OF CONTENTS

CHAPTER NO	TITLE	Pg. No
	ACKNOWLEDGEMENT	V
	ABSTRACT	VI
1.	INTRODUCTION	1
	1.1 INTRODUCTION	1
	1.2 OBJECTIVE OF PROJECT WORK	2
	1.3 ORGANIZATION OF CHAPTERS	
	1.3.1 INTRODUCTION	3
	1.3.2 LITERATURE REVIEW	3
	1.3.3 SYSTEM DESIGN	3
	1.3.4 DESIGN CALCULATION	3
	1.3.5 SPEED AND STRENGTH OF THE ROOF'S MOVEMENT	4
	1.3.6 ASSEMBLY DRAWING	4
	1.3.7 PROCESS CHART	4
	1.3.8 EXPERIMENTATION	5
	1.3.9 COST ESTIMATION	5
	1.3.10 CONCLUSION	5
2.	LITERATURE REVIEW	6
	2.1 INTRODUCTION	
	2.2 LITERATURE SURVEY	6
	2.3 SUMMARY	7

3.	DESIGN CALCULATION	
	3.1 CALCULATION OF BENDING MOMENT OF BASE FRAME	8
	3.2 MOMENT OF INERTIA FOR BASE FRAME	8
	3.3 STRESS ACTING MATERIAL	8
	3.4 GEAR CALCULATION :	8
	3.5 TORQUE CALCULATION AT FIRST LEAD :	9
4.	SYSTEM ARCHITECTURE	
	4.1 DESIGNING AND FABRICATION	12
	4.1.1 TOP VIEW	12
	4.1.2 MOVER	12
	4.1.3 ISOMETRIC VIEW	13
	4.2 SCHEMATICS DIAGRAM OF SMART WATER DISPENSING SYSTEM.	
	4.2.1 SCHEMATIC OF ATMEGA328P MICROCONTROLLER	14
	4.2.2 SCHEMATIC DIAGRAM OF L298N MOTOR DRIVER	14
5.	COMPONENTS OF AUTOMATIC ROOFING SYSTEM	
	5.1 COMPONENTS SPECIFICATION	
	5.1.1 ATMEGA328P MICROCONTROLLER	15
	5.1.2 L298N MOTOR DRIVER	18
	5.1.3 DC GEAR MOTOR	19
	5.1.4 ULTRASONIC SENSOR	20
6.	COST ESTIMATION	
	6.1 ESTIMATION TABLE	
	6.1.1 ADVANTAGES	21
	6.1.2 DISADVANTAGES	22
	6.1.3 APPLICATIONS	23
	6.1.4 CONCLUSION	23
	REFERENCES	26

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In the modern era, the need for sustainable living and efficient resource management has become increasingly important. Growing concerns about water scarcity, rising energy costs, and the need for automation in everyday systems highlight the importance of intelligent solutions. Our project titled “Design and Fabrication of sustainable Smart Water Filling System Using Sensor Based Automation” aims to address these issues by combining automation, monitoring and communication in a single system.

The system is built around a microcontroller that controls and coordinates various sensors and actuators. An ultrasonic sensor monitors water levels to avoid overflow and dry running of the pump, while a temperature sensor ensures proper control of the water heating process. Relays are used to automate the pump and heater, and a GSM module provides real-time alerts to the user about critical conditions. Motor drivers are also included for mechanical operations, and status indicators give a quick view of the system’s functioning. The design ensures that water and energy are used efficiently, minimizing wastage while improving convenience for users. By using an SMPS (Switch-mode power supply) and buck converters, the system is powered safely and reliably with reduced energy loss. This approach ensures that the system can be applied in residential buildings, commercial spaces, and small-scale industries where resource management is a growing concern.

Overall, this project demonstrates how simple electronic components can be integrated to build a practical and affordable solution. It not only addresses immediate needs such as water conservation and energy efficiency but also contributes to the development of smart and sustainable infrastructure for the future.

1.2 OBJECTIVE OF PROJECT WORK

The objective of this project is to design and fabricate an automatic roofing system that autonomously responds to environmental changes, providing protection for outdoor spaces. It aims to integrate sensors (rain, light, and wind) with a control unit to automatically open or close the roof based on real-time weather conditions. The project focuses on using lightweight, durable materials like aluminum or polycarbonate for the roof structure, ensuring energy efficiency and user convenience. The system will include a manual override for added flexibility and be thoroughly tested for reliability and safety in various conditions.

1.3 ORGANIZATION OF CHAPTERS

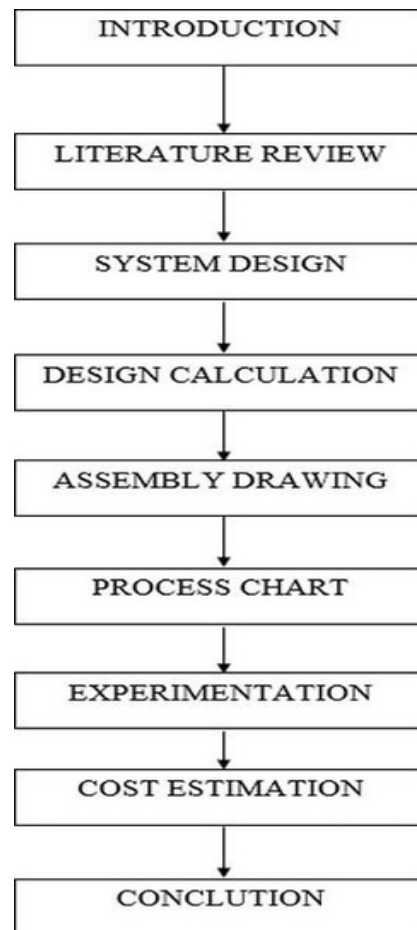


Fig 1.1.1 Organisation of chapter

1.3.1 INTRODUCTION

This section introduces the purpose and motivation behind the project. It explains the need for a roofing system that can automatically detect rain and cover outdoor areas such as terraces, parking spaces, or patios. The introduction discusses about manual override importance, for instance, when the us wants to control the roof in non-rain conditions, for maintenance, or other specific reasons.

1.3.2 LITERATURE REVIEW

This step involves studying existing roofing systems, especially those that use sensors to detect rain and respond automatically. The literature review looks at technologies like water or moisture sensors, actuators for mechanical movement, control circuits, and roofing materials that respond well to weather condition

1.3.3 SYSTEM DESIGN

System design focuses on creating a blueprint for the roofing system. This step involves determining the core components: a rain sensor (to detect rainfall), a motorized mechanism (to move the roof), a control unit (which processes the rain sensor's signals), and a manual override mechanism (like a switch or remote). It specifies how these components work together.

1.3.4 DESIGN CALCULATION

In this stage, precise calculations are done to make sure the system works efficiently. These calculations include:

- o **Water tank capacity:** How large the tank should be, depending on the required storage volume.
- o **Motor load rating:** How powerful the motor must be to lift or circulate water effectively.
- o **Temperature threshold:** Setting up the limit of heat levels that should activate the control system.

1.3.5 SPEED AND STRENGTH OF THE ROOF'S MOVEMENT

Ensuring the devices turn on and off at proper timing without overloading the circuit. These calculations guarantee that the system is not only responsive and automated but also safe, efficient, and reliable.

1.3.6 ASSEMBLY DRAWING

The assembly drawing visually shows how each component will be placed and assembled. This diagram includes:

- o **Placement of sensors:** The best locations to install the rain sensors to ensure accurate detection.
- o **Motor and mechanical linkages:** How the motors and pump are connected to the water system to operate effectively.
- o **Control system layout:** Where the Arduino, GSM module, and wiring will be arranged for reliable operation.
- o **Relay and driver units:** The relays and motor driver module will be positioned for proper switching of pump, heater, and motors.

1.3.7 PROCESS CHART

The process chart outlines the operational flow of the automation system. It shows step-by step how the system functions in real-world scenarios:

- o When water level rises above the set point, the ultrasonic sensor sends a signal to the Arduino.
- o The Arduino processes this signal and activates the pump or motor through the relay and driver.
- o When temperature reaches the threshold, the LM35 sends input, and the heater is controlled automatically.
- o If abnormal readings or critical levels are detected, the GSM module sends an alert message to the user.
- o Manual control can override the system at any point, allowing switching of pump, heater, or motors as required.

1.3.8 EXPERIMENTATION

This stage involves building a prototype or scaled model of the smart monitoring and control system. The experimentation phase is crucial to test:

- o How well the ultrasonic sensors detect the water level and trigger the pump control system.
- o The accuracy of the LM35 sensor in measuring temperature and activating the heater without delay or errors.

1.3.8 COST ESTIMATION

Cost estimation involves calculating all the expenses related to developing and installing the system. It includes:

- o **Material costs:** Sensors, GSM module, relays, motor driver, wiring, pump, heater, and control board.
- o **Labor costs:** The cost of hiring technicians or contractors to assemble and install the system.
- o **Maintenance costs:** Estimating long-term costs for replacing sensors, servicing the GSM module, or repairing relays. This stage helps ensure that the system is reliable and provides a clear budget for stakeholders.

1.3.9 CONCLUSION

The smart monitoring and control system successfully achieved its goals of detecting water levels, measuring temperature, and automatically operating the pump and heater while sending alerts via GSM. It proved to be accurate, reliable, and practical. Future improvements could include cloud-based IoT integration, solar power for energy efficiency, and mobile app control for remote monitoring. Overall, the system offers an effective solution for efficient water and temperature management in both domestic and industrial settings.

CHAPTER - 2

LITERATURE REVIEW

2.1 INTRODUCTION

The literature on automated water filling systems highlights the integration of mechanical design, automation technologies, sensing mechanisms, and user-friendly operation. Researchers have investigated a variety of approaches ranging from smart pumping units, sensor-based monitoring, robotic grippers, to advanced actuation methods for precise control.

Several studies emphasize the role of ultrasonic sensors, weight sensors, and encoders for accurate water level monitoring and refill alerts, ensuring reliability in real-time applications. Similarly, mechanical innovations such as belt drives, telescopic tubes, and cam-based grippers have been explored to achieve precise motion and automation in handling.

In addition, work on temperature-controlled systems ensures water can be maintained at user-preferred conditions, while electronic design advancements such as switch-mode power supplies (SMPS) improve system stability and efficiency. These multidisciplinary contributions demonstrate how automation and mechatronics can transform traditional water dispensing into smart, efficient, and sustainable systems.

2.2 LITERATURE SURVEY :

Developed a self-actuated linear drive for electromechanical systems, achieving smooth and precise motion over a 600 mm range with minimal backlash. This demonstrated how belt drives offer cost-effective and silent motion transmission for automation [1] . Proposed a smart mineral water bottle using IoT that enabled controlled pumping without contamination. The study showed that diaphragm pumps provide precise flow control, making them suitable for dispenser applications [2] .

Designed a six-finger robotic gripper using servo motors and compliant fingers. It successfully grasped irregular objects such as bottles and fruits, showing how CAM profile regulation with spring loading ensures adaptive gripping [3] . Developed an ultrasonic sensor-based water level controller with Arduino for domestic tanks. It effectively prevented overflow and dry running, proving that sensor-based monitoring ensures efficient and consistent tank filling [4].Proposed a smart water dispenser monitoring system using weight sensors and notification modules. The system accurately detected empty cans and sent alerts via SMS, demonstrating that weight sensors are reliable for tracking can usage through Wheatstone bridge circuits [5]. Integrated a telescopic tube with motor actuation using a lead screw mechanism for pick-and place robotics. The system achieved ± 2 mm repeatability, showing that lead screw-driven actuation provides precise vertical extension with high torque[6].

2.2 SUMMARY OF FINDINGS

The literature on automated water dispensing and mechatronic systems highlights the growing importance of integrating sensors, actuators, and control technologies to improve efficiency and reliability in daily-use applications. Studies on ultrasonic sensors, weight sensors, and encoders demonstrate how real-time monitoring ensures accurate water level detection, prevents overflow or shortages, and provides timely alerts for refilling. Mechanical design contributions, such as belt drives, telescopic tubes, and cam-based grippers, illustrate cost effective and precise motion control methods suitable for automation in handling heavy water cans.

Studies emphasize hygienic pump systems, reliable temperature control, and efficient power management for smart water filling systems. These innovations enhance reliability, reduce manual effort, and improve comfort, with future potential in IoT integration, energy optimization, and full automation.

CHAPTER 3

DESIGN CALCULATION

3.1 CALCULATION OF BENDING MOMENT OF BASE FRAME:

$$M = WL^2/8$$

$$W = (160 \times 9.81)/0.8$$

M - BENDING MOMENT

$$= 1962 \text{ N/m}$$

W - WEIGHT

$$M = (1962 \times 0.64)/8$$

$$= 156.96 \text{ N}$$

3.2 MOMENT OF INERTIA FOR BASE FRAME:

$$C = 0.04\text{m}, I = 3.28 \times 10^{-7} \text{ m}^4$$

C - CENTER DISTANCE

I - MOMENT OF INERTIA

3.3 STRESS ACTING MATERIAL:

$$\begin{aligned}\sigma &= M.C/I \\ &= 6.2784/3.28 \times 10^{-7} \\ &= 27.8\text{MPa}\end{aligned}$$

$$\text{Factor of Safety} = 250/27.8 = 8.9 \quad (\text{SAFE - ACCORDING TO 'THE ENGINEERING TOOL BOX'})$$

3.4 GEAR CALCULATION :

- Inner diameter (d) = 19mm
- Outer diameter (D) = 38mm
- Torque to transmit = 0.67Nm
- Pitch/mean diameter (d1) = 28.5mm

Assuming $m=0.8$

$$\sigma_b = 47/7.536 \times 10^{-3} \times 0.1284 \times 3,7696 \times 10$$

$$\sigma_b = 12.88 \text{ MPa}$$

- Module $m = 0.8 \text{ mm}$
- Face width $b = 7.5 \text{ mm}$
- Teeth $Z = 20$
- Inner diameter $d_1 = 19.5 \text{ mm}$
- Addendum diameter $A_d = 21.6 \text{ mm}$
- Pitch diameter $P_d = 20 \text{ mm}$
- Calculated bending stress $\sigma_b = 12.98 \text{ MPa}$

This is well below polycarbonate yield ($\sim 60\text{-}80 \text{ MPa}$). So acceptable.

Convert d_1 to meters: $28.5 \text{ mm} = 0.0285 \text{ m}$

$$F_a = T / d_1 / 2$$

$$F_a = 0.67 / 0.0285 / 2 = 47.02 \text{ N}$$

$$Y = 0.154 - 0.912 / 19 = 0.154 - 0.048 = 0.106$$

LEWIS FORMULA:

$$\sigma_b = F_a / b \times Y \times P_c$$

CONVERT DIMENSIONS TO METERS;

$$b = 0.014137 \text{ m}, P_c = 0.004712 \text{ m}$$

$$\sigma_b = 47.02 / 0.014137 \times 0.106 \times 0.004712$$

3.5 TORQUE CALCULATION AT FIRST LEAD :

$$\tan \pi = \mu$$

$$\pi = \tan^{-1} \mu$$

$$\pi = \tan^{-1} (\mu) = \pi = \tan^{-1} (0.25) = 14.03^\circ$$

$$P \times d / 2 \times R = \text{Gear diameter} \times \text{Force} / 2$$

$$\text{Total torque at pipe circumference} = P \times d / 2$$

$$= 74.05 \times 19.05 / 2$$

$$= (1348.025 \text{ Nmm}) / 2 \times 10^3$$

$$= 0.67 \text{ Nm}$$

CONCLUSION

The base frame design for the smart water filling system ensures structural safety, efficiency, and durability. Through accurate load analysis and proper material selection, the frame can withstand operational loads without deformation or failure. The calculated stress is well below the material's yield strength, indicating a high safety factor. Overall, the design provides a strong, reliable, and cost-effective foundation that enhances the system's stability, sustainability, and long-term performance.

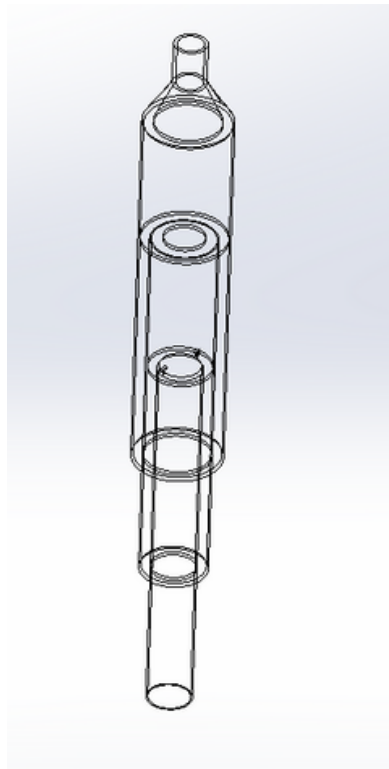


Fig 3.1 telescope tube

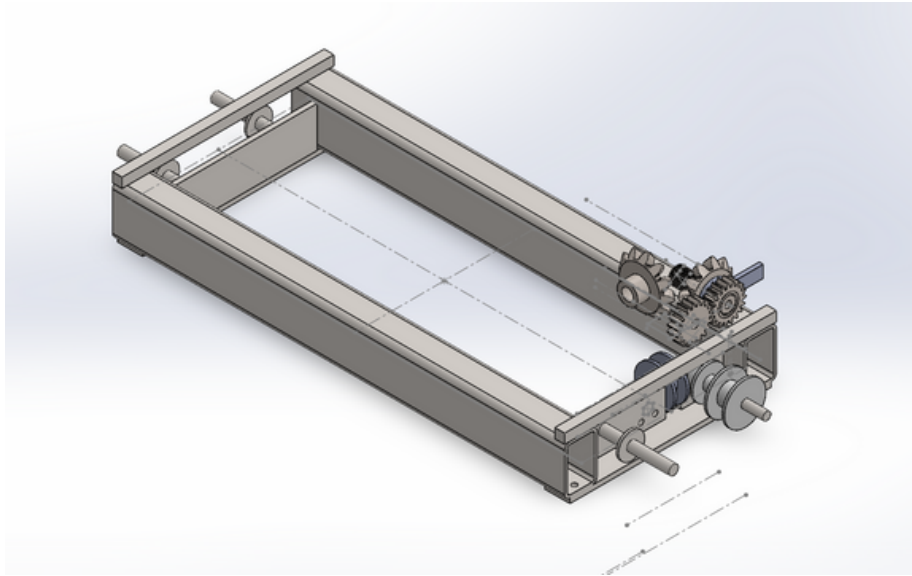


Fig 3.2 Mover with differential assembly

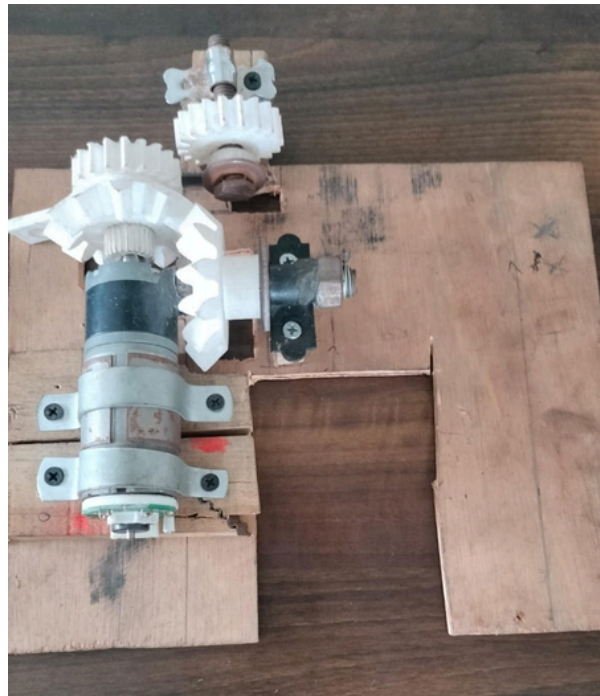


Fig 3.3 Differential assembly

CHAPTER 4

SYSTEM ARCHITECTURE

4.1 DESIGNING AND FABRICATION

4.1.1 TOP VIEW

The top view of the frame shows the can slots, the mover mechanism for telescopic movement, and the telescopic tube, illustrating the component placement and overall system layout.

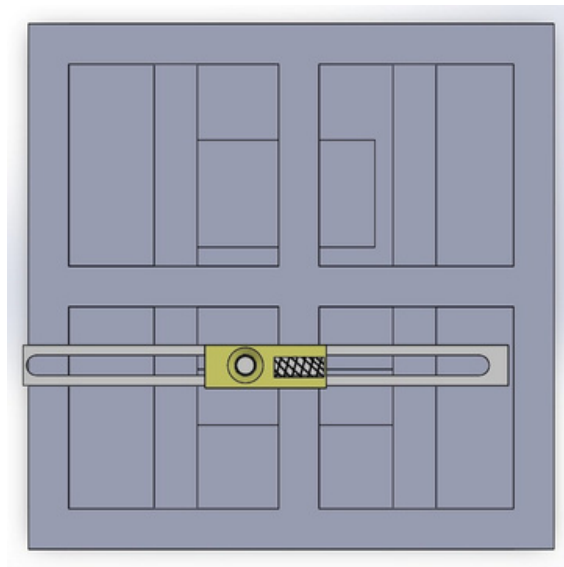


Fig 4.1.1 Top View

4.1.2 MOVER

The front view shows the mover supporting the telescope in its expanded position, illustrating the extension mechanism and alignment.

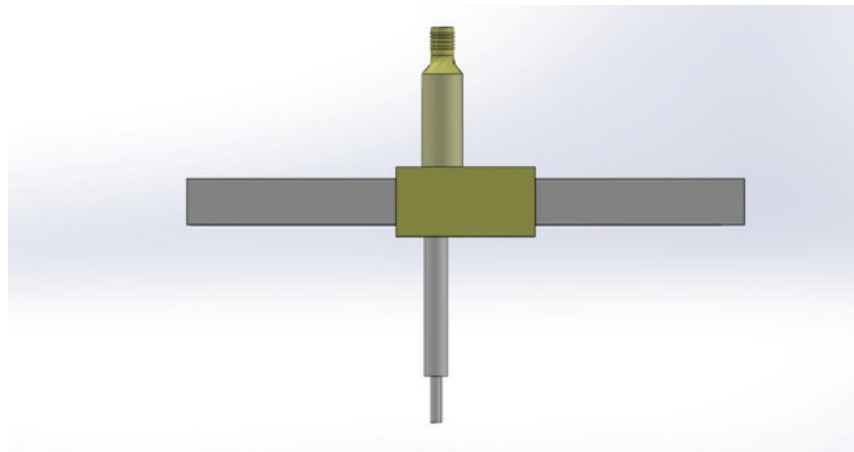


Fig 4.1.2 Mover

4.1.3 ISOMETRIC VIEW:

The isometric view displays the mover, telescopic tube, water pump, and can slots, providing a 3D perspective of the system's overall structure and component arrangement.



Fig 4.1.3 Isometric View

4.2 SCHEMATICS AND CIRCUIT DIAGRAM OF SMART WATER DISPENSING SYSTEM.

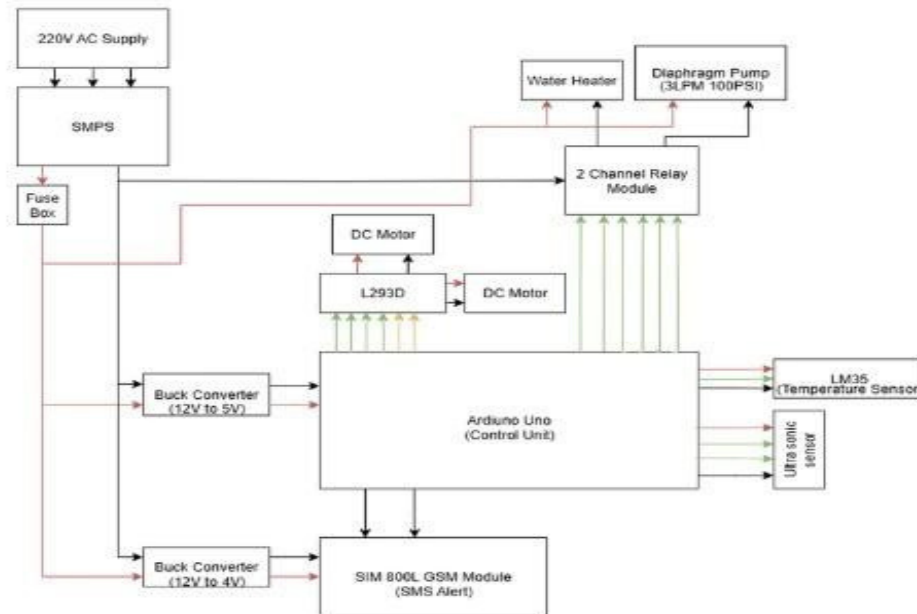


Fig 4.2.1 schematics and circuit diagram

4.2.1 SCHEMATICS OF ATMEGA328P MICROCONTROLLER

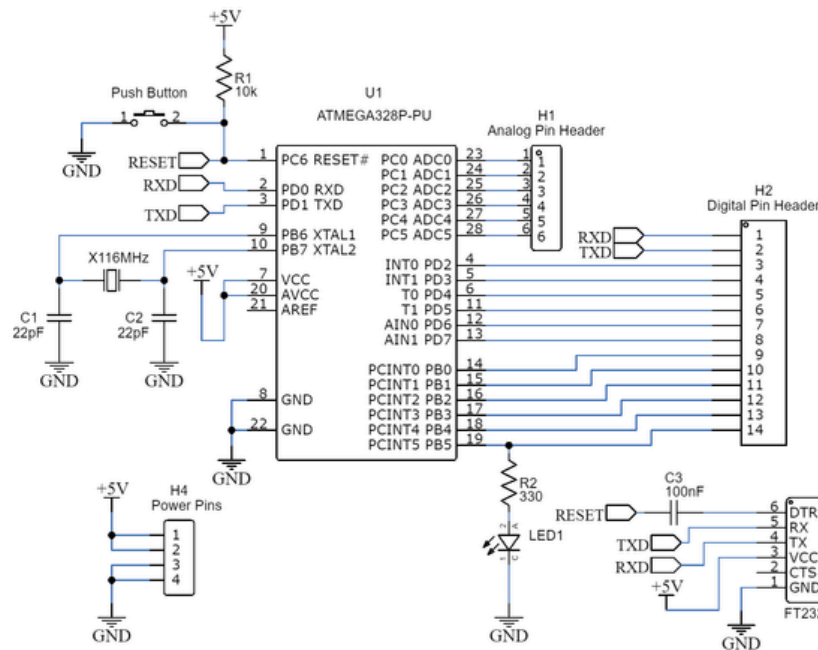


Fig 4.2.2 Atmega328p Microcontroller schematic diagram

4.2.3 SCHEMATIC DIAGRAM OF L298N MOTOR DRIVER

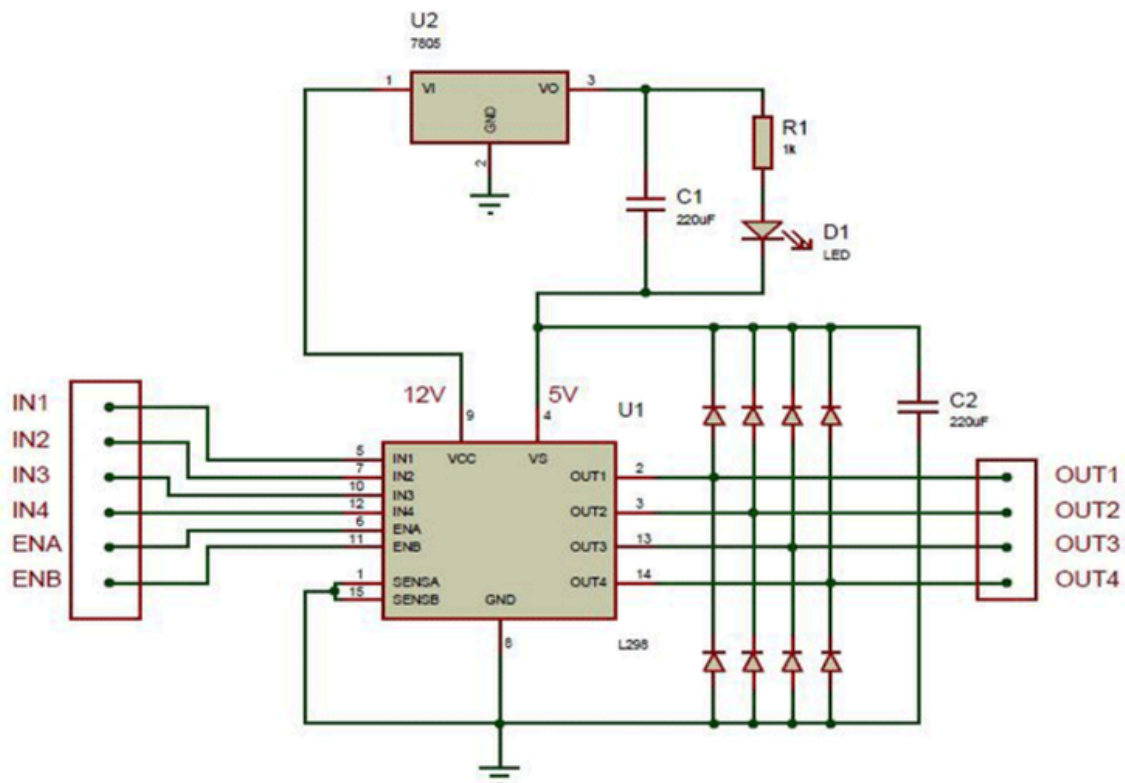


Fig 4.2.3 L298n motor driver schematic diagram

CHAPTER 5

COMPONENTS OF SMART WATER DISPENSING SYSTEM

5.1 COMPONENTS SPECIFICATION:

5.1.1 ATMEGA328P Microcontroller

The ATmega328P is an 8-bit AVR microcontroller widely used in embedded systems and Arduino boards, known for its versatility and low power consumption. It operates at up to 20 MHz, with 32 KB of flash memory, 2 KB of SRAM, and 1 KB of EEPROM. The chip supports 23 GPIO pins, features a 10-bit ADC with six channels, and includes three timers (two 8-bit and one 16-bit) for precise timing and PWM(Pulse-width modulation) control. Communication options include UART, SPI, and I2C(Inter-Integrated Circuit). With its low power modes and bootloader support, it is ideal for sensor interfaces, robotics, and low-power applications.

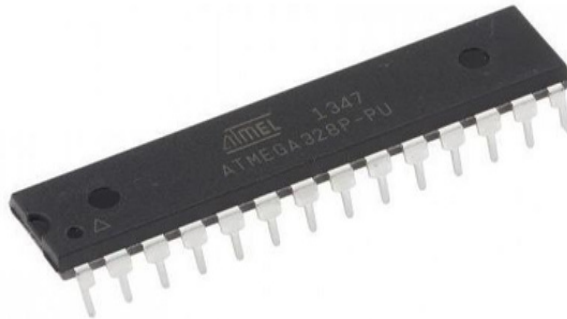


Fig 5.1 Atmega328P microcontroller

Water Level Detection: The Arduino Uno can interface with ultrasonic sensors. By measuring the echo response time, it calculates the water level accurately and uses its digital pins to trigger further actions.

Temperature Monitoring: Using the LM35 temperature sensor, the Uno reads analog signals through its ADC (analog to digital converter). It can monitor temperature variations and take decisions such as activating the water heater when required.

Motor and Pump Control: Through the L298N motor driver and relays, the Arduino Uno controls DC motors and pumps. PWM(pulse width modulation) outputs regulate motor speed, while digital pins manage the relay switching for pump and heater operation.

Decision-Making Logic: Using its programmable flash memory and interrupt system, the Arduino Uno executes control algorithms to automatically activate the pump, motor, or heater based on sensor data.

Communication with External Systems: The Arduino Uno works with the GSM800 module using UART. It can send SMS alerts to users about water level or temperature conditions, ensuring reliable remote monitoring. **Power Management:** With buck converters providing stable 5V supply, the Uno maintains reliable operation. Low-power modes of the ATmega328P can also be leveraged for energy efficiency in extended use.

Override Control: The microcontroller can also provide a manual override function, where users can manually operate the roof using switches, buttons, or remote control, with the ATmega328P handling user input via its GPIO(General purpose input/output) pins.

In this project, the Arduino Uno plays a central role in processing sensor data, controlling actuators, and enabling communication, making the mini project an efficient and automated water and temperature management system.

Table 5.1 specification of Atmega328P microcontroller

SPECIFICATION	DETAILS
CoreArchitecture	8-bitAVR(RISC)
OperatingVoltage	1.8Vto5.5V
MaximumClock Speed	20 MHz
ProgramMemory (Flash)	32 Kb
SRAM	2 KB
EEPROM	1 KB
NumberofI/OPins	23 GPIO pins
ADC	10-bit, 6 channels 6
PWM Channels	UART, SPI, 12C
CommunicationInterfaces	ExternalOscillator
PowerConsumption	1 mA

In conclusion, the ATmega328P microcontroller stands out as a highly versatile, reliable, and cost-effective solution for a wide range of embedded system applications. Its combination of affordability, ease of use, and broad community support ensures that it will remain a popular choice for embedded systems development for years to come. As the heart of popular platforms like Arduino, the ATmega328P has gained recognition for its balance of performance, power efficiency, and ease of use. With its 8-bit architecture, 32KB of flash memory, and 2KB of SRAM, it is well-suited for a variety of projects that require moderate computational power without consuming.

5.1.2 L298N MOTOR DRIVER

The L298N motor driver controls the direction and speed of DC motors in automated systems. It uses input signals to change motor direction and PWM to adjust speed. In an automated Water filling system, it enables smooth, precise moving of telescopic tube.

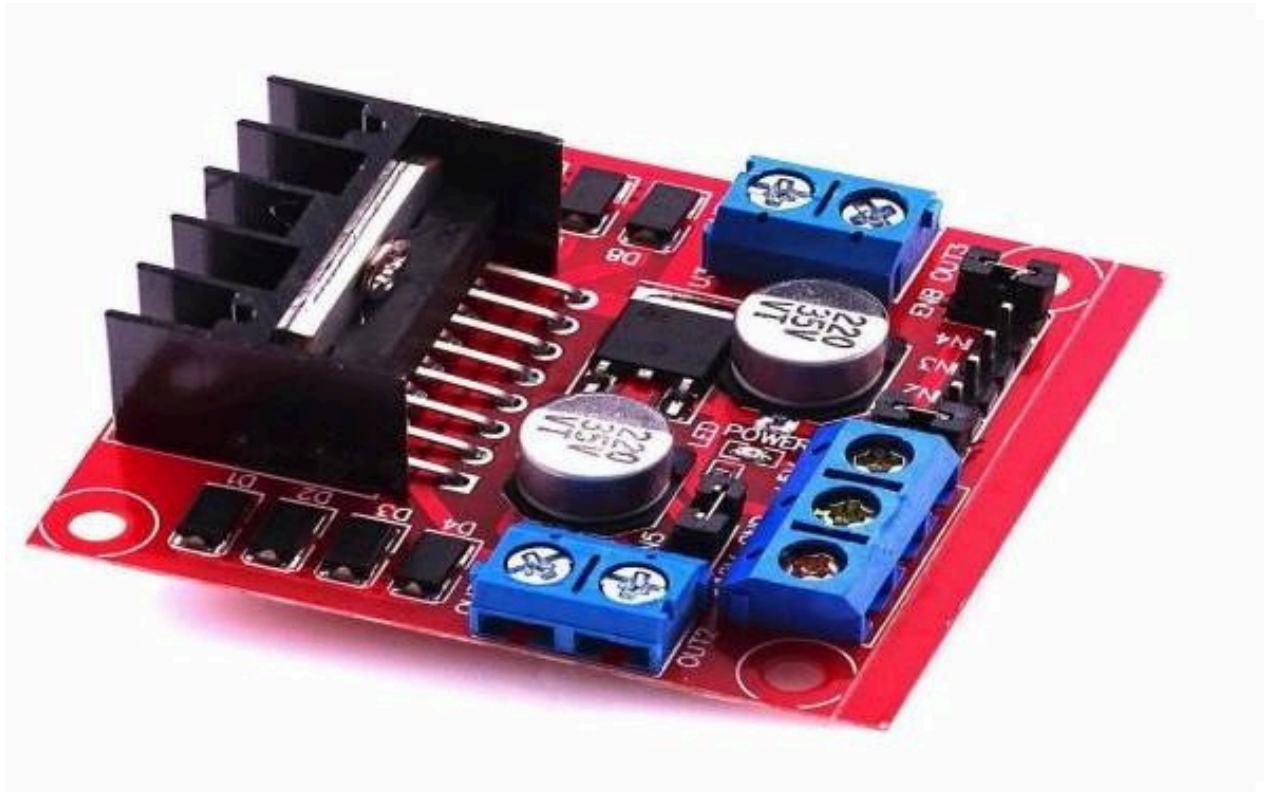


Fig 5.1.2 L298n motor driver

The L298N motor driver is essential for controlling DC motors in an automated water filling system. It allows the ATmega328P to manage motor direction and speed, using PWM signals. The L298N can control two motors, ideal for precise moving of telescopic tube, and operates at a voltage range of 5V to 35V, with a maximum current of 2A per channel. It has built-in over-temperature protection, ensuring reliable operation in varying conditions.

5.1.3 DC GEAR MOTOR

In the smart water filling system, the DC gear motor plays a crucial role in moving the telescopic tube, providing the necessary torque for precise positioning during the filling process. When an empty can or bottle is detected, the ATmega328P microcontroller sends control signals to the L298N motor driver, which regulates the motor's speed and direction. The high torque of the gear motor ensures smooth and reliable movement, preventing sudden jerks or misalignment that could affect filling accuracy.



Fig 5.1.3 DC gear motor

A DC gear motor is a compact motor integrated with a gear mechanism to transmit motion efficiently, making it ideal for applications requiring precise control. It operates on direct current (DC) and uses the gear assembly to adjust the speed-to-torque ratio. This makes it perfect for automation systems like the smart water filling setup, where controlled, powerful, and accurate motion is essential. The gear motor ensures smooth operation even under varying loads, and its speed can be easily adjusted using pulse-width modulation (PWM) signals for optimized performance.

5.1.4 ULTRASONIC SENSOR



Fig 5.1.4 Ultra Sonic sensor

Ultrasonic sensors are key components in automation systems, offering non-contact, fast, and accurate distance or level measurements. They work by emitting sound pulses and measuring the time it takes for the echo to return, allowing precise calculation of the distance to an object. In the smart water filling system, these sensors reliably monitor water levels and trigger actions such as operating motors, pumps, or relays based on set thresholds. Their use enhances accuracy, safety, and real-time control, making the system more efficient and convenient.

CHAPTER 6

COST ESTIMATION

6.1 ESTIMATION TABLE:

The components are selected and the price has been finalized from the vendors, overall cost will come around.

Table 6.1 estimation table

SNO	Parts	Quantity	COST(RS)
1	Frame	1	1700
2	Reservoir Tank	1	250
3	$\frac{3}{4}$ inch hose	1	200
4	Atmega 328P	1	400
5	5V DC Motor	1	1000
6	SMPS Unit	1	1000
7	Diaphragm Water Pump 12v	1	400
8	Temperature sensor	1	150
9	Ultrasonic Sensor	2	50
10	Telescopic tube	1	200
11	Stepper Motor	1	400
12	Timing Belt	3m	500
13	Gear	6	400
14	GSM Module	1	400
15	Wire	As required	200
		Total	7250

CHAPTER 6

ADVANTAGES AND DISADVANTAGES & APPLICATIONS

ADVANTAGES:

- **Automation:** Automatically monitors water level, temperature, and rainfall, reducing the need for manual supervision.
- **Cost-Effectiveness:** Built with affordable components like Arduino, sensors, and GSM modules, offering a low-cost alternative to industrial controllers.
- **Scalability:** Can be expanded with additional sensors or actuators for domestic, agricultural, or industrial applications.
- **Remote Monitoring:** GSM module sends instant SMS alerts about water levels, temperature, or abnormal conditions for real-time awareness.
- **Energy Efficiency:** Pumps and heaters operate only when needed, saving electricity and reducing energy waste.
- **Resource Management:** Conserves water and electricity by controlling devices based on actual requirements.
- **Safety:** Reduces risks of overheating, water overflow, or dry running, protecting equipment.
- **User-Friendly:** Easy to configure and operate, requiring minimal technical knowledge, suitable for households, farms, and small industries.

DISADVANTAGES :

- **Cost:** Although cheaper than large industrial systems, the combined cost of sensors, GSM modules, and drivers can be higher than simple manual setups; upgrading or replacing components in large-scale applications adds to expenses.
- **Maintenance:** Sensors and relays require periodic calibration, cleaning, or replacement; troubleshooting wiring faults or GSM connectivity issues may need technical expertise.
- **Power Dependency:** The system depends on a continuous power supply; frequent outages can interrupt operation unless a backup source is provided.
- **Environmental Limitations:** Ultrasonic and other sensors may be affected by dust, noise, or extreme weather, potentially causing inaccurate readings or false triggers.

APPLICATIONS

Water Management: The system can control water tanks in households, farms, and industries by monitoring water levels and automatically switching pumps on or off, preventing overflow and ensuring supply.

Temperature Control: By monitoring real-time temperature, it can regulate heaters in domestic water systems, aquaculture, or agricultural setups, ensuring optimal conditions without manual intervention.

Agriculture: Farmers can benefit from automated irrigation and water-level monitoring, reducing manual effort while conserving water and electricity in remote areas. **Industrial Use:** In small-scale industries, the system can maintain process water levels, prevent pump damage, and send alerts to supervisors in case of abnormal conditions.

Disaster Prevention: The rain sensor and GSM alerts can notify users about sudden rain or water rise, helping in flood-prone areas to take early safety measures.

Smart Homes: In smart homes, the integration of automated roofing systems with home automation technology provides unmatched convenience and control. Integrated with home automation systems, automatic roofs can be controlled remotely or through sensors, providing convenience and energy savings by controlling sunlight and ventilation.

CONCLUSION

The adoption of an automated water management and control system marks a transformative leap in modern household and industrial applications. This advanced solution not only enhances convenience but also provides users with intelligent monitoring of essential resources. With features integrated into microcontroller-based systems, users can effortlessly track water levels, temperature, and system status in real-time, while also receiving instant alerts via GSM modules for quick decision-making and improved safety. Beyond convenience, automated water management systems promote energy efficiency and sustainability, contributing to lower electricity usage and responsible water conservation. Their ability to control pumps and heaters automatically reduces wastage, prevents overflows, and supports eco-friendly practices, making them a valuable asset for resource conscious communities.

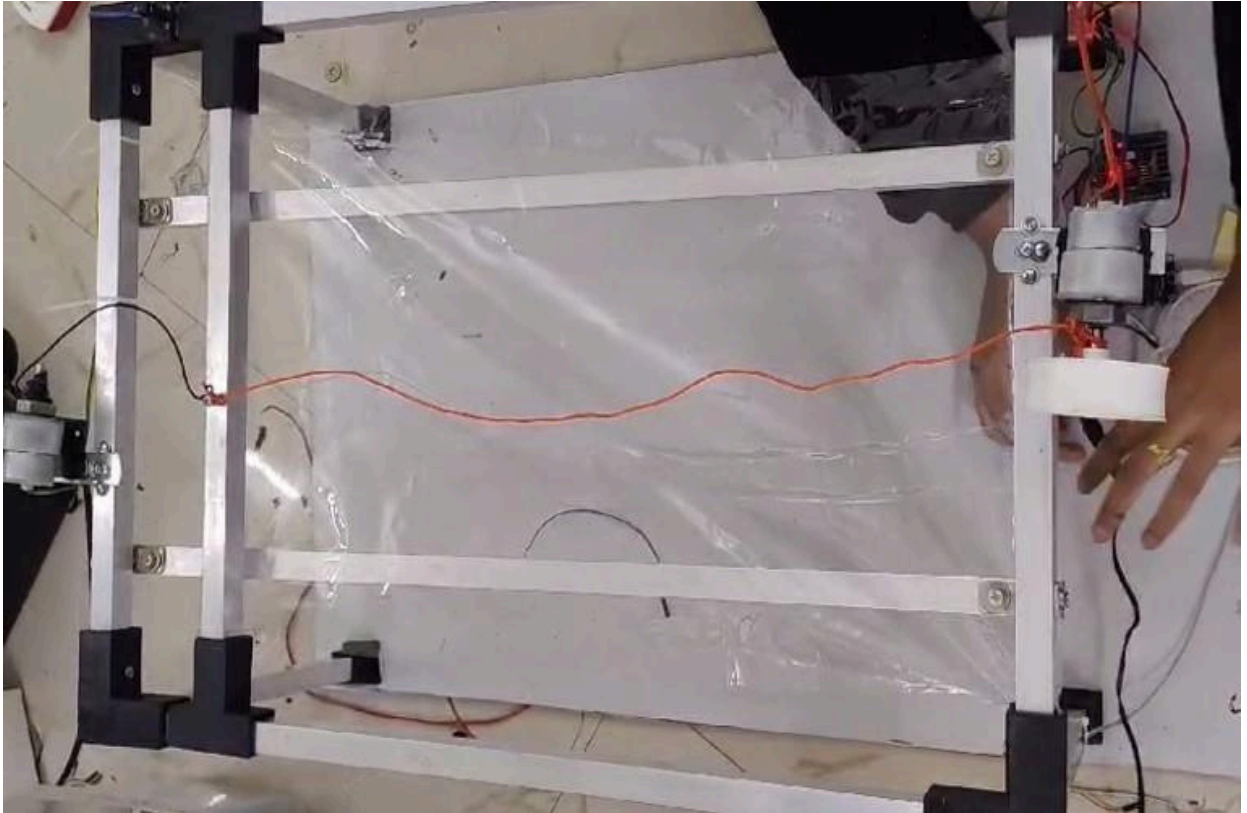


Fig no 6.2 Project Outcomes

While the initial setup may require additional components and calibration compared to conventional manual systems, the long-term benefits—including reduced manual effort, enhanced safety, and minimized resource loss—far outweigh the costs. As advancements in embedded technology continue to emerge, such systems are expected to evolve further with improved sensors, wireless connectivity, and smarter automation features, enriching both domestic and industrial applications.

In summary, automated water and temperature control systems represent not just a functional upgrade but a visionary step toward intelligent infrastructure.

By regulating pumps, monitoring heat levels, and sending instant alerts during abnormal conditions, these systems help reduce reliance on manual supervision while lowering risks of equipment failure or accidents.

As embedded systems and IoT-based solutions continue to advance, integration with renewable technologies and AI-based analytics could further enhance efficiency, offering predictive maintenance and optimized usage patterns. With increasing

awareness of resource management and safety, the demand for such automated solutions is expected to rise steadily, establishing them as an integral part of sustainable development initiatives.

Automated water management systems are a powerful tool that combines embedded technology with safety, convenience, and efficiency. They empower users with a level of control over resources that was previously unavailable, offering personalized monitoring while reducing wastage, energy consumption, and operational risks.

Whether it's ensuring adequate water supply, preventing equipment damage, or delivering real-time updates to the user, these systems represent a smart investment in the future of automation. As innovation progresses, such projects will likely become an even more integral part of smart environments, contributing to a more sustainable, safe, and efficient future for both homes and industries.

REFERENCES

- a) **V.B. Bhandari** *Design of Machine Elements*, McGraw-Hill Education. Provides fundamentals of designing mechanical parts such as shafts, couplings, and actuators used in the dispenser frame and pump mechanism.
- b) **J.L. Meriam and L.G. Kraige** *Engineering Mechanics: Dynamics*, Wiley. Covers force and motion analysis, essential for understanding motor-driven actuation of telescopic tubes and the handling of water cans.
- c) **Max Rabiee** *Programmable Logic Controllers (PLCs): Hardware and Programming*, Cengage Learning. Explains the basics of automation and control, relevant for microcontroller/PLC-based water level monitoring and pump actuation.
- d) **Mikell P. Groover** *Automation, Production Systems, and Computer-Integrated Manufacturing*, Pearson. Provides insights into integrating automation, robotics, and control systems into real world applications like automated dispensing.
- e) **William D. Callister and David G. Rethwisch** *Materials Science and Engineering: An Introduction*, Wiley. Useful for selecting food-grade, corrosion-resistant materials for water storage tanks and dispenser frames.
- f) **Austin Hughes** *Electric Motors and Drives: Fundamentals, Types and Applications*, Newnes. Explains selection and application of motors for pumping systems, telescopic motion, and robotic gripper actuation.
- g) **Robert L. Norton** *Machine Design: An Integrated Approach*, Pearson. Offers advanced insights into design of mechanical linkages and actuators used in William D. Callister and David G. Rethwisch, *Materials Science and Engineering: An Introduction*, robotic gripping mechanisms.
- h) **Ernest Oberg et al.** *Machinery's Handbook*, gripper actuation., Industrial Press. A classic reference for mechanical design data, tolerances, and standards useful in fabricating the tray and telescopic units.